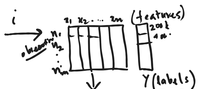


$y_{price} = f(x)$

features

Supervised

Unsupervised

Reinforcement

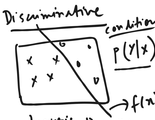
Agent $\leftrightarrow$ environment

Reward / Penalty

Discriminative

conditional probability

$P(Y|X)$



logistic Regression

Generative model

$P(X, Y)$

Discriminative Models

Linear Model

$$f(x) = w_1 x_1 + w_2 x_2 \dots w_n x_n \to y$$

$$y = w_1 x_1 + w_2 x_2 + \dots w_n x_n$$

(prix) $y = w^T x$ ← OLS

Linear Regression

Loss $\to$ MSE $\to$ $\min \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2$

$\hat{y} = w^T x$

w^* (optimal) $= (X^T X)^{-1} X^T y$

MLE (estimate $\theta(w)$)

$\hat{\theta} = \underset{\theta}{\operatorname{argmin}}\ L(\theta) \to$ Loss

$\hat{\theta}_{MLE} = \underset{\theta}{\operatorname{argmax}}\ P(D|\theta)$

Assumption: IID

$$P(D|\theta) = \prod_{i=1}^{N} P(y_n|x_n, \theta)$$

$$= \sum_{i=1}^{N} \log P(y_n|x_n, \theta)$$

$$\hat{\theta}_{MLE} \overset{\operatorname{argmax}}{=} \sum_{n=1}^{N} \log P(y_n|x_n, \theta)$$

$$NLL = -\operatorname{argmin} \sum_{n=1}^{N} \log P(y_n|x_n, \theta)$$

min (NLL)

$\to$ MAP

MLE $\to$ Bayes Theorem

$$P(Y|X) = \frac{P(X|Y)\,P(Y)}{\text{evidence}}$$

MLE for Bernoulli

(PDF) $P(y_n|\theta) = \theta^{y_n} (1-\theta)^{1-y_n}$

$$NLL = -\log \sum_{n=1}^{N} \theta^{y_n} (1-\theta)^{1-y_n}$$

$$= -N_1 \log \theta + N_0 \log(1-\theta)$$

$N = N_1 + N_0$

$$\frac{d(NLL)}{d\theta} = -\frac{N_1}{\theta} + \frac{N_0}{1-\theta} \xrightarrow{\text{set } 0}$$

$$\hat{\theta}_{MLE} = \frac{N_1}{N_0 + N_1}$$

MLE for Normal distribution

$NLL(\mu, \sigma^2)$ $\quad \frac{\partial(NLL)}{\partial \mu} \xrightarrow{\text{set } 0}$

$\frac{\partial(NLL)}{\partial(\sigma^2)} \xrightarrow{\text{set } 0}$ solve for σ^2

Logistic Regression $\to$ classification

$\hat{y} = w^T x \to$

$\hat{y} = \sigma(w^T x)$

$$\sigma(a) = \frac{1}{1+e^{-a}}$$

$$\sigma'(a) = \sigma(a) \cdot (1 - \sigma(a))$$

$$\sigma(-a) = 1 - \sigma(a)$$

MLE

$$P(y|x, \theta) = \mathrm{Ber}(y | \sigma(w^T x))$$

$$NLL = -\frac{1}{N} \sum_{i=1}^{N} \log \left[\mu_n^{y_n} (1-\mu_n)^{1-y_n} \right]$$

where $\mu_n = \sigma(w^T x)$, θ

$$= -\frac{1}{N} \left(y_n \log \mu_n + (1-y_n) \log(1-\mu_n) \right)$$

(cross-entropy) $\quad$ min (NLL)

Loss function $\to$ optimization problem

$\downarrow$ solve

Gradient Descent

Loss